

Pre-work · Module 4 Integration

All nine Part A chapters. This is a synthesis sheet, so do not read anything new for it. Work from the four packets you have already built and from your own mini visuals. Every chunk here crosses at least two chapters, and the crossings are where the module is actually tested.

Module 4 · 5 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first.
This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

This packet cuts across the chapters instead of following them. Eight chunks, each built from at least two systems. If a chunk sends you back to the reading rather than back to your own earlier visuals, that is the material to repair first.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 GRID

Cortex against medulla, four organs

Organ down the side: kidney, adrenal gland, ovary, lymph node. What sits in the cortex and what sits in the medulla across the top. Star the one organ whose two regions are built from entirely different tissues.

2 GRID

The four tubes of the trunk

System down the side: respiratory, digestive, urinary, reproductive. What it carries, where it begins, where it opens to the outside, and the sphincter or valve that guards it across the top. Circle the two rows that share a final passage in the male.

3 CHAIN

One drop of dietary fat, villus to bloodstream

A chain that crosses two chapters. Start in a lacteal inside a villus, then run through the lymphatic collecting vessels, a node, a trunk, the cisterna chyli, the thoracic duct, and into the subclavian vein. Write beside the first box that this is the one absorbed thing that never entered a blood capillary in the intestinal wall.

4 GRID

Organs with two jobs

Organ down the side: pancreas, ovary, testis, liver, kidney, pharynx, male urethra, placenta. The two jobs across the top, and a third column naming the tissue or region that does each one. This grid is the reason the same organ turns up on two different sheets.

DRAWING 1

The trunk in one sagittal section, four systems at once

BRAIN DUMP FIRST

Three minutes. Every tube of Module 4 in one body, in the right order front to back.

Draw the trunk cut down the midline, from the nasal cavity to the pelvic floor, with the diaphragm across the middle. Lay in the airway in front and the alimentary canal behind it, and mark the one region where the two share a passage. Carry the canal on through the stomach, the intestines, and the rectum. Behind all of that draw one kidney, its ureter, the bladder, and the urethra. Then add the pelvic reproductive organs for one sex and mark the point where their duct joins or crosses the urinary tract.



In your notebook, head the page **M4-1 The trunk in one sagittal section, four systems at once** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at any level of your section and name every tube crossing it, front to back, without a label.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A section shows an organ with a capsule, an outer region packed with rounded clusters of cells, and a plainer inner region. Several vessels enter through the capsule and a single vessel leaves at an indentation on one side. Name the organ, and say what rules out the kidney and the ovary.

Answer. A lymph node.

Reasoning. Read the vessels first, because a cortex and a medulla alone fit four organs in this module. Many vessels entering through the capsule and one leaving at a hilum is the traffic pattern of a lymph node, where afferent vessels arrive all around the surface and a single efferent vessel exits at the notch. The kidney also has a cortex, a medulla, and a hilum, but every vessel it owns enters and leaves at that hilum rather than through the capsule, and its medulla is built of cone-shaped pyramids, not a plain inner region. The ovary carries rounded structures in its cortex too, its follicles, but nothing is filtered through it, and its vessels sit in the medulla instead of crossing the capsule. The rounded clusters in the cortex of a node are lymphocytes, and they are exactly why a node reads like an ovary at low power. When two organs share a plan, let the vessels decide.

Set A · Name it

1. The vessels that carry lymph into a node are the _____.
2. Organs such as the kidneys, pancreas, duodenum, and parts of the colon, which lie behind the peritoneum, are described as _____.
3. The one region of the pharynx that carries air only is the _____.
4. The gland that caps the superior pole of each kidney is the _____.
5. Name the four organs in this module that have both a cortex and a medulla.

5 HUB

The shelf behind the peritoneum

The posterior abdominal wall at the hub. Spoke to every structure Part A places behind the peritoneum: both kidneys, the pancreas, the duodenum, and the parts of the colon. Then add the adrenal gland sitting on top of each kidney and the ureter running down from each one, and write beside every spoke which packet it came from.

6 SPLIT

Exocrine against endocrine

One split. Duct or no duct, where the product is delivered, and how far it acts. Hang three Module 4 examples off each branch, then write the pancreas across the joint between them, because it is both at once.

7 HUB

The same trick in four places, surface area

Write increase the surface at the hub. Spoke to the alveoli of the lung, the circular folds with villi and microvilli of the small intestine, the million nephrons of one kidney, and the follicles of the thyroid gland. Beside each spoke write what has to cross or be stored on that surface.

8 LADDER

Every sphincter and valve, in order

One ordered list down the page, in the order a structure appears along the tubes: upper and lower esophageal, pyloric, sphincter of Oddi, ileocecal, internal and external anal, then internal and external urethral. Beside each write which tube it guards, and circle the two you can control.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

Four organs with a cortex and a medulla

BRAIN DUMP FIRST

Two minutes. Four sections side by side, and what fills each region of each one.

Draw four small sections in a row: a kidney, an adrenal gland, an ovary, and a lymph node. Label the capsule, the cortex, and the medulla on each. Then fill each region with what belongs in it: renal corpuscles and convoluted tubules in the renal cortex and pyramids in the renal medulla, three zones in the adrenal cortex and modified nervous tissue in its medulla, follicles in the ovarian cortex and vessels in its medulla, and clustered lymphocytes in the cortex of the node with its hilum on the medullary side.

In your notebook, head the page **M4-2 Four organs with a cortex and a medulla** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can be shown any one of the four in section and name the organ from the contents of its cortex alone.

Set B - Predict it

1. A tube in the trunk is cut and a clear watery fluid drains from it rather than blood. Predict which system it belongs to and explain how you narrowed it down.

2. A single organ is removed and both a digestive secretion and a hormone are lost. Name the organ and predict both losses.

3. The skeletal muscles of the trunk and limbs are inactive for a long period. Predict the effect on lymph return and explain why breathing still helps.

4. A surgeon operating on the anterior neck must spare four glands the size of grains. Name them and predict what their position puts at risk.

Set C - Tell it apart

1. Three organs in this module are wrapped by a serous membrane. Name them, name each membrane, and say what the space between the two layers is for.

2. Distinguish an exocrine gland from an endocrine gland, then explain why the pancreas is listed as both.

3. The alveolus, the villus, and the renal corpuscle all solve the same problem. State the problem and compare the three solutions.

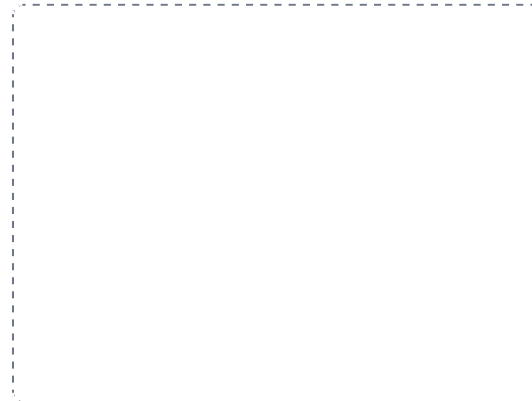
DRAWING 3

Three exchange surfaces, side by side

BRAIN DUMP FIRST

Two minutes. Three barriers drawn at the same scale, with every layer named.

Draw three enlarged details in a row. First an alveolus against a pulmonary capillary, with the alveolar epithelium, the fused basement membranes, and the capillary endothelium of the respiratory membrane. Second one intestinal villus, with its absorptive cells and their microvilli, the capillary bed inside it, and the lacteal down the middle. Third a renal corpuscle, with the fenestrated endothelium, the basement membrane, and the filtration slits between the podocyte processes. Under each one write what crosses and what is held back.



In your notebook, head the page **M4-3 Three exchange surfaces, side by side** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say for each of the three what stays behind, and why a different barrier is built at each site.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Cortex against medulla, four organs

GRID

2 The four tubes of the trunk

GRID

3 One drop of dietary fat, villus to bloodstream

CHAIN

4 Organs with two jobs

GRID

5 The shelf behind the peritoneum

HUB

6 Exocrine against endocrine

SPLIT

7 The same trick in four places, surface area

HUB

A large rectangular area defined by a dashed line, intended for the student's answer to question 6.A large rectangular area defined by a dashed line, intended for the student's answer to question 7.

8 Every sphincter and valve, in order

LADDER

A large rectangular area defined by a dashed line, intended for the student's answer to question 8.